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Week 3

SESSION-5

Task 1: To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

Code:

```
UW PICO 5.09 File: 25.c Modified

{
    printf("Error: No tokens found.\n");
    return 1;
}

/* Display tokens */
display_tokens(tokens, token_count);

/* Validate */
printf("=====\n");
printf("      TOKEN VALIDATION\n");
printf("=====\n");

validate_tokens(tokens, token_count);

printf("\nParsing completed successfully.\n");

return 0;
}

Get Help      WriteOut      Read File      Prev Pg      Cut Text      Cur Pos
Exit          Justify        Where is       Next Pg      UnCut Text    To Spell
```

Output:

```
rohmiller@macBook-Air: ~ % gcc 25.c -o 25
rohmiller@macBook-Air: ~ % ./25
=====
SINGLE QUOTE PARSER
=====

Example:
'hello world'
some string
'this is B0DER'
abc 'hello world' xyz

Enter command: ped

=====
PARSING RESULTS
=====
token 0
Type : WORD
Value : [ped]

=====
TOKEN VALIDATION
=====
Validation: PASSED

Parsing completed successfully.
rohmiller@macBook-Air: ~ %
```

Task 2: To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

Code:

```
UW PIC0 8.09 File: 26.c Modified

/* Display tokens */
display_tokens(tokens, token_count);

/* Validate */
printf("=====\\n");
printf("      OUTPUT VALIDATION\\n");
printf("=====\\n");
if (validate_tokens(tokens, token_count))
{
    printf("\\nQuoted command parsed successfully.\\n");
}
return 0;
}

Get Help  WriteOut  Read File  Prev Pg  Out Text  Cur Pos
Exit      Justify  Where is  Next Pg  OnOut Text To Spell
```

```
Last login: Sat Aug 29 09:31:34 on ttys000
nrx@nallas-mac-Book-Air ~ % nano 26.c

1

nrx@nallas-mac-Book-Air ~ % gcc 26.c -o 26
nrx@nallas-mac-Book-Air ~ % ./26
=====
DOUBLE QUOTE PARSER
=====
Examples:
"hello world"
"Hello $USER"
echo "Home is $HOME"
"User: $(USER)"
Enter command: 
```

Session 6:

Task 1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

Code:

```
UW PICO 5.09 File: 27.c Modified

/* Display tokens */
display_tokens(tokens, token_count);

/* Validate */
printf("\n=====n");
printf("      OUTPUT VALIDATION\n");
printf("=====n");

if (validate_tokens(tokens, token_count))
{
    printf("\nEscape parsing completed successfully.\n");
}

return 0;
}

^G Get Help      ^G WriteOut      ^R Read File      ^Y Prev Pg      ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify        ^W Where is       ^V Next Pg      ^U UnCut Text    ^T To Spell
```

Output:

```
mrk@hallas-mac-Book-Air ~ % gcc 27.c -o 27
mrk@hallas-mac-Book-Air ~ % ./27
=====
ESCAPE SEQUENCE PARSER
=====

Supported escape sequences:
\ -> space
\\ -> \
\' -> '
\" -> "
\0 -> \0
\n -> \n

Examples:
hello\world
hello\"world
\\home
file\\name

Enter command: pwd

=====
PARSER OUTPUT
=====
Token 1 : (pwd)

=====
OUTPUT VALIDATION
=====
Validation: PASSED

Escape parsing completed successfully.
mrk@hallas-mac-Book-Air ~ %
```

Task 2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

Code:

```
UW PICO 5.89 File: 38.c Modified

    }
    printf("Command execution failed.\n");
}
else if (WIFSIGNALED(status))
{
    printf("Child was terminated by signal %d.\n",
        WTERMSIG(status));
}

printf("\nParent process continues.\n");
printf("Program completed successfully.\n");
}

return 0;
}

Get Help      WriteOut    Read File    Prev Pg      Out Text    Our Pos
Exit          Justify     Where is     Next Pg      UnOut Text  To Spell
```

```
nrk@nallas-mac-Book-Air ~ % gcc 30.c -o 30
```

```
nrk@nallas-mac-Book-Air ~ % ./30
```

```
=====
COMMAND LAUNCHER USING execvp()
=====
```

```
Enter a command with arguments.
Example: ls -l
Example: echo Hello World
Example: pwd
```

```
Command> pwd
```

```
Arguments received: 1
Argument 1: pwd
```

```
-----
PARENT PROCESS
```

```
Parent PID : 2904
Child PID  : 2905
```

```
Parent is waiting for child...
```

```
-----
CHILD PROCESS
```

```
Child PID : 2905
Parent PID : 2904
Executing  : pwd
```

```
/Users/nrk
```

```
Child process finished.
Child exited normally.
Exit status: 0
Command executed successfully.
```

```
Parent process continues.
Program completed successfully.
nrk@nallas-mac-Book-Air ~ %
```